

The Hong Kong Polytechnic University

Department of Electronic and Information Engineering

EIE2111 Lab 6: Arrays and Vectors

(Deadline for Submission: Check the course information)

Introduction

This laboratory exercise is designed to give you hand-on experience in Arrays and Vectors.

Questions

1. A small airline has just purchased a computer for its new automated reservation system. You have been asked to program the new system. You write a program to assign seats on each flight of the airline (capacity: 6 seats). Your program should display the following menu —Please type '0' for reservation, type '1' for checking and type '2' for exit. If the person types 0, your program should prompt the user to input his/her name, the type of seats (first class or economy class) and the number of seats. Use a one-dimensional array of objects to represent the seating chart of the plane. A class should be provided for creating objects in the array. The class should include two parameters: the passenger's name (a string) and an integer to indicate whether a seat is occupied or not. Initialize all the elements (objects) of the array to indicate that all seats are empty. When a seat is assigned, set the corresponding integer to 1 to indicate that the seat is no longer available and also the passenger's name to reserve such seat. Your program should, of course, never assign a seat that has already been assigned. When the first class section is full, your program should ask the person if it is acceptable to be placed in the economy section (and vice versa). If yes, then make the appropriate seat assignment. If no, then print the message "Next flight leaves in 3 hours."

The following is an example to create an array of objects:

```
class abc
{
    ...
}

int main()
{
    abc arrayOfObjects[10];
    ...
}
```

You may have a problem if you use both `cin` and `getline` to read the input. If you use `cin` first and then `getline`, you need to put a dummy `getline` between them to remove the newline character from `cin`.

2. Write a program that simulates the rolling of two dices. The program should use `rand` to roll the first dice and should use `rand` again to roll the second dice. The sum of the two values should then be calculated. [Note: Each dice can show an integer value between 1 and 6, so the sum of the two values will vary from 2 to 12, with 7 being the most frequent sum and 2 and 12 being the least frequent sums.] Note that there are 36 possible combinations of the two dices. Your program should roll the two dices 3,600 times. Use a one-dimensional array to store the frequencies of the sums of the two dices. Print the results in a tabular format. Also, determine if the totals are reasonable (i.e., there are six ways to roll a 7, so approximately one-sixth of all the rolls should be 7).
3. Combine the above two programs into one. The following is the sample output:

```

C:\Windows\system32\cmd.exe
Question 1
Reservation <Press '0'>, Check <Press '1'>, or Exit <Press '2'>? 0
Please input the following information
Name: Lawrence Cheung
First class <Press '1'> or Economy class <Press '0'>? 1
Number of seats: 2
Reserved. 2 seats are first class.

Reservation <Press '0'>, Check <Press '1'>, or Exit <Press '2'>? 1
First class:
Seat 1 : Lawrence Cheung
Seat 2 : Lawrence Cheung
Economy class:

Reservation <Press '0'>, Check <Press '1'>, or Exit <Press '2'>? 0
Please input the following information
Name: Helen Wong
First class <Press '1'> or Economy class <Press '0'>? 1
Number of seats: 2
Not enough seats! Do you accept some seats are economy class?
(Press '1' to accept or '0' to reject) 1
Reserved. 1 seats are first class and 1 seats are economy class.

Reservation <Press '0'>, Check <Press '1'>, or Exit <Press '2'>? 1
First class:
Seat 1 : Lawrence Cheung
Seat 2 : Lawrence Cheung
Seat 3 : Helen Wong
Economy class:
Seat 1 : Helen Wong

Reservation <Press '0'>, Check <Press '1'>, or Exit <Press '2'>? 0
Please input the following information
Name: Jack Ho
First class <Press '1'> or Economy class <Press '0'>? 0
Number of seats: 3
Not enough seats!
Next flight leaves in 3 hours.

Reservation <Press '0'>, Check <Press '1'>, or Exit <Press '2'>? 2

Question 2
Please enter the seed: 2
  Sum      Total Expected   Actual
    2         95   2.778%    2.639%
    3        243   5.556%    6.750%
    4        276   8.333%    7.667%
    5        402  11.111%   11.167%
    6        496  13.889%   13.778%
    7        609  16.667%   16.917%
    8        459  13.889%   12.750%
    9        378  11.111%   10.500%
   10        318   8.333%    8.833%
   11        211   5.556%    5.861%
   12        113   2.778%    3.139%

Press any key to continue . . .

```

Instructions

- a. You are required to submit your C++ programs (the whole projects created in Microsoft Visual Studio 2019) in Question 3 to Blackboard. Zip all of them into a single file.
- b. The deadline of the submission: **Check the course information.**
- c. It is not required to create any header files for the above exercises. It is fine if all program codes are in the main program.

Lawrence Cheung
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